Code: 23AME05

R23

II B.Tech II Semester Regular Examinations, June - 2025

Optimization Techniques

(Common to CSE(AIML), CSE(DS), IT & CSE(AI))

(For 2023 Admitted Batch Only)

Time: 3 Hours

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Max.Marks:70

PART-A

Answer all questions and all questions carry equal marks 10 x 2 = 20M

1.a) Write the feasibility condition for simplex method.
1.b) What is meant by linear programming?
1.c) List the methods of obtaining initial basic feasible solution of a transportation problem.
1.d) Define an unbalanced transportation problem.
1.e) Write the assumptions in sequencing problem.
1.f) Write the Johnson's algorithm for solving n jobs on 2 M/C's.
1.g) Write the characteristics of game theory.
1.h) Write maximin – minimax principle.
1.i) Define Total float, Free Float and independent float.
1.j) Write the assumptions in PERT.

PART-B

Answer All questions and all questions carry equal marks 5 x 10 = 50M

2 a) Solve the following LPP using Simplex Method.

Minimize $Z = 8x_1 - 2x_2$

S/ to $-4x_1 + 2x_2 \leq 1$

$5x_1 - 4x_2 \leq 3$

and $x_1, x_2 \geq 0$

(OR)

b) Solve the following LPP using Two - Phase Method.

Minimize $Z = 4x_1 + x_2$

S/ to $3x_1 + x_2 = 3$

$4x_1 + 3x_2 \geq 6$

$x_1 + 2x_2 \leq 4$

and $x_1, x_2 \geq 0$

3 a) Find the non- degenerative basic feasible solution for the following transportation problem using.

(i) North – west corner method
(ii) Least cost method
(iii) Vogel's approximation method

	A	B	C	D	Supply
P	10	20	5	7	10
Q	13	9	12	8	20
R	4	5	7	9	30
S	14	7	1	0	40
T	3	12	5	19	50
Demand	60	60	20	10	

(OR)

b) A company has four M/C's to do three jobs each job can be assigned to one and only one M/C. The cost of each job on each M/C is given in the following table, assign jobs.

	1	2	3	4
A	18	24	28	32
B	8	13	17	19
C	10	15	19	22

4 a) Solve the following sequencing problem giving an optimal solution if passing is not allowed.

	M1	M2	M3	M4
A	13	8	7	14
B	12	6	8	19
C	9	7	8	15
D	8	5	6	15

(OR)

b) Compute the minimum total elapsed time needed to process jobs 1 and 2 on five machines A, B, C, D and E given the following data:

Job1	Sequence	A	B	C	D	E
	Time (hrs)	3	4	2	6	2
Job2	Sequence	B	C	A	D	E
	Time (hrs)	5	4	3	2	6

5 a) Is the following two – person Zero – sum game stable? Solve the game

	I	II	III	IV
1	5	-10	9	0
2	6	7	8	1
3	8	7	15	1
4	3	4	-1	4

(OR)

b) Solve the rectangular game whose payoff matrix is

-1	-2	8
7	5	-1
6	0	-12

6 a) Draw the network diagram and find EST, EFT, LST, LFT, TF, FF, IF, project duration and critical path.

Activity	1-2	1-3	1-5	2-3	2-4	3-4	3-5	3-6	4-6	5-6
Duration	8	7	12	4	10	3	5	10	7	4

(OR)

b) Construct the network for the project whose activities and the time estimates of the activities (in weeks). Compute (i) Expected duration of each activity t_e (ii)Expected variance of each activity σ^2 (iii)Expected variance of the project length.

Activity	1-2	2-3	2-4	3-5	4-5	4-6	5-7	6-7	7-8	7-9	8-10	9-10
t_o	3	1	2	3	1	3	4	6	2	1	4	3
t_m	4	2	3	4	3	5	5	7	4	2	6	5
t_p	5	3	4	5	5	7	6	8	6	3	8	7

Code: 23AHS21

R23

II B.Tech II Semester Regular Examinations, June - 2025

Probability & Statistics

(Common to CSE, CSE(AIML), IT, CSE(AI), CSE(IOT) & CSBS)

(For 2023 Admitted Batch Only)

Time: 3 Hours

*****

Max.Marks:70

PART-A

Answer all questions and all questions carry equal marks 10 x 2 = 20M

1.a) What is the range of the correlation coefficient?

1.b) Define population and sample.

1.c) Define Continuous random variable with example.

1.d) State addition theorem of Probability.

1.e) Write any two properties of Poisson distribution

1.f) Define Normal distribution

1.g) Explain Type I and Type II errors.

1.h) Write the test static of single proportion for large samples.

1.i) What are the applications of student t- test.

1.j) Write any two properties of - F-test.

PART-B

Answer All questions and all questions carry equal marks 5 x 10 = 50M

2 a) Find the mode of the following distribution

Class interval	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	5	8	7	12	28	20	10	10

(OR)

b) Obtain the rank correlation coefficient for the following data

X	68	64	75	50	64	80	75	40	55	64
Y	62	58	68	45	81	60	68	48	50	70

3 a) The continuous probability function $f(x) = kx^2e^{-x}$ when $x \geq 0$. Find (i) k (ii) Mean (iii) Variance.

(OR)

b) In a bolt factory machines A, B, C manufactures 20%, 30%, and 50% of the total of their output and 6%,3% and 2% are defective. A bolt is drawn at random and found to be defective. Find the probabilities that it is manufactured from (i) Machine A (ii)Machine B (iii) Machine C

P.T.O

4 a) Fit the Binomial distribution to the following data

x	0	1	2	3	4	5
f	2	14	20	34	22	8

(OR)

b) If X is a Poisson Variate such that $P(X = 0) = [P(X = 2) + 3P(X = 4)]$ find (i) Mean (ii) $p(x \leq 2)$

5 a) A sample of 11 rats from a central population had an average blood viscosity of 3.92 with a standard deviation of 0.61. Estimate the 95% confidence limits for the mean blood viscosity of the population.

(OR)

b) The means of two large samples of sizes 1000 and 2000 members are 67.5 inches and 68 inches respectively. Can the samples are regarded as drawn from the same population of standard deviation 2.5 inches.

6 a) Two independent samples of 8 and 7 items respectively had the following values. Is the difference between the means of sample significant?

Sample I	11	11	13	11	15	9	12	14
Sample II	9	11	10	13	9	8	10	-

(OR)

b) Given the following contingency table for hair color and eye color. Is there good association between the two?

		Hair color			
		Fair	Brown	Black	Total
Eye color	Blue	15	5	20	40
	Grey	20	10	20	50
	Brown	25	15	20	60
	Total	60	30	60	150

R23

Code: 23ACA02

II B.Tech II Semester Regular Examinations, June - 2025

Machine Learning

(CSE(AIML) & CSE(AI))

(For 2023 Admitted Batch Only)

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Time: 3 Hours　　　　　　　　　　　　　　　　　　　　　　Max.Marks:70

PART-A

Answer all questions and all questions carry equal marks　　　　$10 \times 2 = 20M$

1.a) What is meant by cross-validation and resampling?
1.b) compare learning by rote and learning by induction?
1.c) What are proximity measures in machine learning?
1.d) Mention few applications of KNN regression.
1.e) State Bayes Rule and its significance in classification.
1.f) What is the role of random forests in classification?
1.g) Compare and contrast Linear Regression and Logistic Regression?
1.h) What is the purpose of the Kernel Trick in Support Vector Machines (SVM)?
1.i) Differentiate between Agglomerative and Divisive Clustering?
1.j) How does Fuzzy C-Means Clustering work?

PART-B

Answer All questions and all questions carry equal marks　　　　$5 \times 10 = 50M$

2 a) Explain the evolution of Machine Learning and its significance in modern computing.

(OR)

b) Explain the concept of Model Learning and its importance. Discuss different algorithms used for model learning in Machine Learning.

3 a) Explain different distance measures used in machine learning. Compare Euclidean distance and Manhattan distance.

(OR)

b) Describe the Radius Distance Nearest Neighbour Algorithm. How does it differ from the standard KNN algorithm?

4 a) Describe the working of Random Forests and how they handle both classification and regression

(OR)

b) Explain the Bayes Classifier, including Bayes Rule and its application in classification.

5 a) Describe the working of the perceptron classifier and its learning algorithm

(OR)

b) Discuss the working of Support Vector Machines (SVM), with an explanation of both the linearly separable and non-linearly separable cases.

6 a) How Matrix factorization works in PCA. Explain in detail?

(OR)

b) Discuss Rough Clustering and Rough K-Means Clustering Algorithm with suitable examples and applications.

R23

Code: 23ACS12

II B.Tech II Semester Regular Examinations, June - 2025

Database Management Systems

(Common to CSE, CSE(AIML), CSE(DS), CSE(AI), CSE(CS), CSBS & IT)

(For 2023 Admitted Batches Only)

*****

Time: 3 Hours | Max.Marks:70

PART-A

Answer all questions and all questions carry equal marks 10 x 2 = 20M

1.a) Define database management system.
1.b) What is an entity relationship model?
1.c) Analyze the characteristics that distinguish the union operation with intersection operation in relational algebra.
1.d) Write the syntax of SELECT statement with example.
1.e) Define Super Key
1.f) What is the difference between a tuple and an attribute?
1.g) List the anomalies of 1NF.
1.h) List the properties of decomposition
1.i) What are the different modes of lock?
1.j) What are different types of schedules are acceptable for recoverability

PART-B

Answer All questions and all questions carry equal marks 5 x 10 = 50M

2 a) Discuss the main characteristics of the database approach and how it is differs from file processing system

(OR)

b) Draw an ER diagram for an application to maintain employee, department, project and dependent details of an employee. Choose the attributes for each entity and make reasonable assumptions about the relations between entities.

3 a) What are the different types of Relation Algebra Operators? Explain in detail

(OR)

b) Consider the following schema for institute library: Student (RollNo, Name, Father_ Name, Branch) Book (ISBN, Title, Author, Publisher) Issue (RollNo, ISBN, Date-of –Issue)
Write the following queries in SQL and relational algebra:
(i) List roll number and name of all students of the branch 'CSE'.
(ii) Find the name of student who has issued a book published by 'ABC' publisher.
(iii) List title of all books and their authors issued to a student 'RAM'.
(iv) List title of all books issued on or before December 1, 2020.
(v) List all books published by publisher 'ABC'

4 a) Consider the following schema for employee database: emp (eno , ename, bdate, title, salary, dno) proj (pno , pname, budget, dno) dept (dno , dname, mgreno) workson (eno , pno, resp, hours)
Write the following queries in SQL and relational algebra:
(i) Write an SQL query that returns the project number and name for projects with a budget greater than $100,000.
(ii) Write an SQL query that returns all works on records where hours worked is less than 10 and the responsibility is 'Manager'.
(iii) Write an SQL query that returns the employees (number and name only) who have a title of 'EE' or 'SA' and make more than $35,000.
(iv) Write an SQL query that returns the departments (all fields) ordered by ascending department name.
(v) Write an SQL query that returns the employee name, department name, and employee title.

(OR)

b) Write SQL query syntax and examples for the following:

(i) To create a table using necessary constraints (Primary Key, NOT NULL, UNIQUE, Check).

(ii) To create a view from a table by bringing few records of a table.

(iii) To alter the table by adding a default value.

(iv) To count the number of records in a table.

(v) To drop the foreign key constraint from the table,

5 a) Describe the different Normal forms of database? Explain the steps to normalize database up to BCNF ?

(OR)

b) Explain briefly insertion, updation and deletion anomalies in databases.

6 a) Elaborate how deadlock is prevented in detail.

(OR)

b) Explain the Two-Phase Locking protocol and its variants.

Code: **23AEC06** **R23**

II B.Tech II Semester Regular Examinations, June - 2025

Digital Logic and Computer Organization

(Common to CSE(AIML), CSE(DS) & CSE(AI))

(For 2023 Admitted Batch Only)

Time: 3 Hours ***** Max.Marks:70

PART-A

Answer all questions and all questions carry equal marks $10 \times 2 = 20M$

1.a) What is Floating point representation?

1.b) Convert $(A9)_{16}$ to binary and decimal?

1.c) Define a Register. Give one application?

1.d) What is the Von Neumann architecture?

1.e) What is the role of the control unit in instruction execution?

1.f) Define multiple-bus organization?

1.g) What is cache hit and cache miss?

1.h) Differentiate between SRAM and DRAM?

1.i) What is polling in I/O systems?

1.j) List any two standard I/O interfaces?

PART-B

Answer All questions and all questions carry equal marks $5 \times 10 = 50M$

2 a) Explain the procedure for converting binary numbers to hexadecimal and vice versa with examples?

(OR)

b) Design a combinational circuit for a 2-bit magnitude comparator and explain its working?

3 a) Describe the working of a JK flip-flop and explain how it can be used to build a counter?

(OR)

b) Explain the basic operational concepts of a computer with a neat block diagram?

4 a) Describe the process of signed-operand multiplication with an example?

(OR)

b) Compare hardwired control and microprogrammed control. Give their advantages and disadvantages?

5 a) Explain the performance considerations in cache memory. How is cache mapped to main memory?

(OR)

b) Discuss secondary storage technologies and their roles in memory hierarchy?

6 a) Explain the structure and working of an interrupt handling mechanism with a diagram?

(OR)

b) Describe standard I/O interfaces and their relevance in modern computer systems?

Code: 23AMB05

R23

II B.Tech II Semester Regular Examinations, June - 2025

Design Thinking & Innovation

(Common to All Branches)

(For 2023 Admitted Batch Only)

Time: 3 Hours

*****

Max.Marks:70

PART-A

Answer all questions and all questions carry equal marks 10 x 2 = 20M

1.a) Identify the principles of design.
1.b) Identify the elements of design and briefly explain their role in design.
1.c) Explain the concept of journey mapping.
1.d) Briefly explain the importance of empathy in understanding user needs
1.e) Define innovation and briefly explain its significance.
1.f) Briefly explain the role of creativity in driving innovation within organizations.
1.g) Define problem formation in the context of product design.
1.h) List the key steps involved in product design.
1.i) List the key principles of Design Thinking
1.j) List ways in which startups can benefit from Design Thinking.

PART-B

Answer All questions and all questions carry equal marks 5 x 10 = 50M

2 a) Explain the role of design elements and principles in shaping successful products, citing examples from various industries.

(OR)

b) Define design thinking and outline its historical evolution.

3 a) Explain the five stages of the design thinking process (Empathize, Define, Ideate, Prototype, Test) and how they contribute to a human-centered solution

(OR)

b) Explain the role of design thinking in fostering social innovations and addressing community challenges.

4 a) Explain the concepts of innovation and creativity, and also how each contributes uniquely to the success of an organization.

(OR)

b) Explain the role of creativity in organizations. Also, explain the processes by which creative ideas are transformed into tangible innovations (creativity to innovation).

5 a) Explain the fundamentals of product design and its role in addressing real-world challenges.

(OR)

b) Explain the key steps involved in product planning and how effective planning contributes to successful product development.

6 a) Explain the core principles of design thinking and how they help redefine business models and decision-making processes.

(OR)

b) Explain how startups can leverage design thinking to develop innovative business solutions and gain a competitive edge in the market.